Computational Modeling

BMED365: Topic List Z01–Z10

BMED365

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Overview

1. **Foundations of Computational Modeling**
   - Z01: Role of mathematical models in biomedicine
   - Z02: Mechanistic vs. phenomenological models
   - Z03: Ordinary differential equations (ODEs) in biology
2. **Neurophysiology Models**
   - Z04: Hodgkin-Huxley model of action potentials
3. **Tumor and Vascular Models**
   - Z05: Tumor growth models
   - Z06: Angiogenesis modeling
4. **Cardiovascular and Muscle Models**
   - Z07: Cardiovascular flow and hemodynamics
   - Z08: ECG and blood pressure waveforms
   - Z09: Muscle force generation models
5. **Cellular and Molecular Models**
   - Z10: Cell signaling and gene regulatory networks

Z01: Explain the role of mathematical models in biomedicine

Mathematical modeling = **using mathematics to describe biological systems**

Definition

A mathematical model is a **simplified representation** of a biological system using equations, allowing us to understand, predict, and simulate behavior.

Why model biological systems?

- **Understanding:** Formalize hypotheses about mechanisms
- **Prediction:** Forecast outcomes (disease progression, treatment response)
- **Experimentation:** Test scenarios that are difficult/unethical in vivo
- **Optimization:** Find optimal treatment strategies

Biomedical applications

Drug dosing, tumor growth prediction, cardiac simulation, neural activity modeling, epidemic forecasting, personalized medicine.

Z02: Distinguish between mechanistic and phenomenological models

Mechanistic models:

- Based on **underlying mechanisms**
- Describe *how* the system works
- Parameters have physical meaning
- Example: Ion channel dynamics

Advantages:

- Interpretable, generalizable
- Can extrapolate beyond data

Disadvantages:

- Complex, many parameters
- Require detailed knowledge

Phenomenological models:

- Describe **observed behavior**
- Focus on *what* happens, not how
- Parameters may lack physical meaning
- Example: Tumor growth curves

Advantages:

- Simple, fewer parameters
- Easier to fit to data

Disadvantages:

- Limited extrapolation
- Less insight into mechanisms

In practice

Often use **hybrid approaches** – mechanistic where knowledge exists, phenomenological where it doesn't.

Z03: Describe ordinary differential equations (ODEs) in biological systems

Ordinary Differential Equations (ODEs):

Definition

ODEs describe how quantities **change over time**:

$$\frac{dx}{dt} = f(x, t)$$

where x is the state variable and t is time.

Biological examples:

Population growth:

$$\frac{dN}{dt} = rN$$

Exponential growth with rate r

Drug elimination:

Enzyme kinetics:

$$\frac{d[S]}{dt} = -\frac{V_{max}[S]}{K_m + [S]}$$

Michaelis-Menten

Predator-prey:

Z04: Explain the Hodgkin-Huxley model of action potentials

Hodgkin-Huxley model (1952, Nobel Prize 1963):

The model

Describes the **ionic basis of action potentials** using ODEs for membrane voltage and ion channel gating:

$$C_m \frac{dV}{dt} = I_{ext} - g_{Na} m^3 h (V - E_{Na}) - g_K n^4 (V - E_K) - g_L (V - E_L)$$

Key components:

- V: Membrane potential
- C_m: Membrane capacitance
- g_{Na}, g_K, g_L: Conductances
- E_{Na}, E_K, E_L: Reversal potentials
- m, h, n: Gating variables (0–1)
- m: Na^+ activation
- h: Na^+ inactivation
- n: K^+ activation

Legacy

Foundation for computational neuroscience. Still used in detailed neuron models and cardiac cell electrophysiology.

Z05: Describe tumor growth models

Mathematical models of tumor growth:

Exponential:

$$\frac{dN}{dt} = rN$$

- Unlimited growth
- Early-stage tumors
- Unrealistic long-term

Logistic:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

- Carrying capacity K
- Resource limitation
- S-shaped curve

Gompertz:

$$\frac{dN}{dt} = rN \ln\left(\frac{K}{N}\right)$$

- Decelerating growth
- Fits clinical data well
- Common in oncology

Clinical applications

Tumor growth models are used to predict progression, plan treatment timing, and evaluate therapy response. Personalized models can incorporate patient-specific parameters.

Z06: Explain angiogenesis modeling and its clinical relevance

Angiogenesis = formation of new blood vessels

Why model angiogenesis?

- Tumors require blood supply to grow beyond ∼1–2 mm
- Angiogenesis is a key therapeutic target (anti-VEGF drugs)
- Models help predict treatment response

Modeling approaches:

Continuum models:

- PDEs for vessel density
- VEGF diffusion and consumption
- Computationally efficient

Discrete/agent-based:

- Individual vessel tips as agents
- Stochastic branching, migration
- More realistic morphology

Clinical relevance

Models predict response to anti-angiogenic therapy (e.g., bevacizumab), help optimize dosing, and explain resistance mechanisms.

Z07: Describe cardiovascular flow models and hemodynamics

Hemodynamics = study of blood flow

Modeling approaches:

0D (Lumped parameter):

- Electrical circuit analogy
- Resistance, compliance, inertance
- Windkessel models

1D models:

- Wave propagation in arteries
- Pulse wave velocity
- Arterial network models

3D CFD:

- Navier-Stokes equations
- Patient-specific geometries
- Wall shear stress, flow patterns

Applications:

- Aneurysm rupture risk
- Stent design
- Surgical planning

Key equation: Poiseuille's law

$Q = \frac{\pi r^4 \Delta P}{8 \mu L}$ (flow rate depends on radius to the 4th power!)

Z08: Understand the relationship between models and ECG/blood pressure

Physiological signals from models:

ECG modeling:

- Bidomain model: Electrical propagation in cardiac tissue
- Cell models (e.g., ten Tusscher) for action potentials
- Forward problem: Compute ECG from heart activity

Applications:

- Arrhythmia simulation
- Drug safety testing (QT prolongation)

Blood pressure waveforms:

- Windkessel models relate cardiac output to arterial pressure
- Pulse wave analysis
- Central vs. peripheral pressure

Applications:

- Arterial stiffness estimation
- Cardiac function assessment
- Virtual patient trials

Digital twins

Patient-specific computational models ("digital twins") can simulate individual responses to interventions.

Z09: Describe muscle force generation models

Muscle contraction models:

Hill's muscle model (classic):

- Three-element model: Contractile element (CE), series elastic (SE), parallel elastic (PE)
- Force-velocity relationship: $F = F_0 \cdot \frac{v_{max} - v}{v_{max} + v/a}$
- Phenomenological, widely used in biomechanics

Cross-bridge models (mechanistic):

- Based on actin-myosin interactions
- Huxley (1957) kinetic model
- States: attached, detached, force-generating
- Calcium activation dynamics
- Captures force-length-velocity relations
- Used in detailed cardiac models

Applications

Movement simulation, prosthetics design, cardiac mechanics, sports biomechanics, rehabilitation.

Z10: Explain cell signaling pathways and gene regulatory networks

Modeling cell signaling and gene regulation:

Cell signaling pathways:

- Cascades of molecular interactions (receptors $\rightarrow$ kinases $\rightarrow$ transcription factors)
- ODE models for concentration dynamics
- Example: MAPK/ERK pathway, PI3K/AKT/mTOR

Gene regulatory networks:

- Genes regulate each other's expression
- Boolean networks: ON/OFF states
- ODE models: Continuous concentrations
- Network motifs: Feedback loops, feedforward
- Emergent behavior: Oscillations, bistability
- Stochastic effects in single cells

Systems biology

Integrating signaling, metabolism, and gene regulation into **whole-cell models** is a major goal of systems biology.

Summary: Computational Modeling

Key points:

- **Z01:** Mathematical models help understand, predict, and simulate biological systems
- **Z02:** Mechanistic models describe mechanisms; phenomenological describe behavior
- **Z03:** ODEs are the workhorse of dynamic biological modeling
- **Z04:** Hodgkin-Huxley: ionic basis of action potentials
- **Z05:** Tumor growth: exponential, logistic, Gompertz models
- **Z06:** Angiogenesis models predict response to anti-VEGF therapy
- **Z07:** Cardiovascular models: 0D lumped parameter to 3D CFD
- **Z08:** ECG/BP waveforms arise from electrophysiology and hemodynamics
- **Z09:** Muscle models: Hill's phenomenological, cross-bridge mechanistic
- **Z10:** Cell signaling and gene networks: systems biology perspective

Key message

Computational modeling bridges theory and experiment in biomedicine. From neurons to tumors to hearts, mathematical models provide quantitative frameworks for understanding disease and optimizing treatment.